Insert Interval (medium)

Problem Statement

Given a list of non-overlapping intervals sorted by their start time, **insert a given interval at the correct position** and merge all necessary intervals to produce a list that has only mutually exclusive intervals.

Example 1:

```
Input: Intervals=[[1,3], [5,7], [8,12]], New Interval=[4,6]
Output: [[1,3], [4,7], [8,12]]
Explanation: After insertion, since [4,6] overlaps with [5,7], we merged them into one [4,7].
```

Example 2:

```
Input: Intervals=[[1,3], [5,7], [8,12]], New Interval=[4,10]
Output: [[1,3], [4,12]]
Explanation: After insertion, since [4,10] overlaps with [5,7] & [8,12], we merged them into [4,12].
```

Example 3:

```
Input: Intervals=[[2,3],[5,7]], New Interval=[1,4]
Output: [[1,4], [5,7]]
Explanation: After insertion, since [1,4] overlaps with [2,3], we merged them into one [1,4].
```

Try it yourself

Try solving this question here:

Java

```java
import java.util.*;

/*class Interval {
  int start;
  int end;

  public Interval(int start, int end) {
    this.start = start;
    this.end = end;
  }
};*/

class Solution {

  public List<Interval> insert(List<Interval> intervals, Interval newInterval) {
    List<Interval> mergedIntervals = new ArrayList<>();
    // TODO: Write your code here
    return mergedIntervals;
  }
}
```

Test Case | Run Code Result | Stdout

```
[[1,3], [5,7], [8,12]]
[4,6]
```

Use Example Testcases | Run | Submit